

Insurance Fraud Detection - Executive Summary

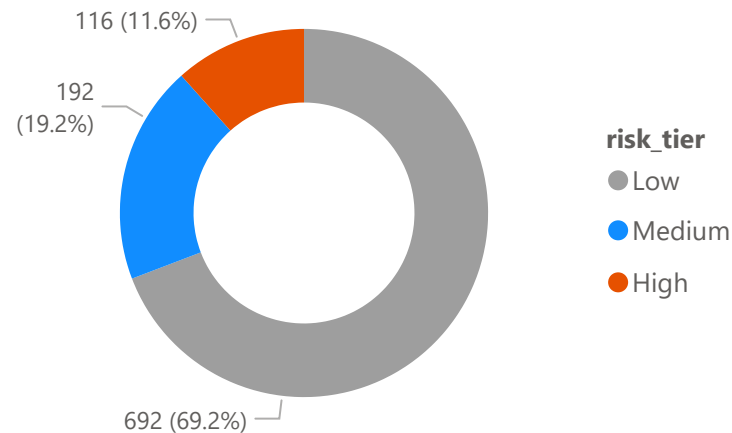
1K
Total Claims

247
Fraud Confirmed

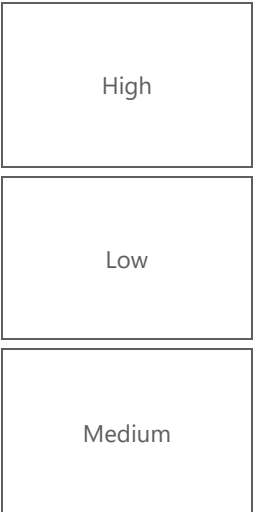
116
High Risk Count

71%
Model Recall

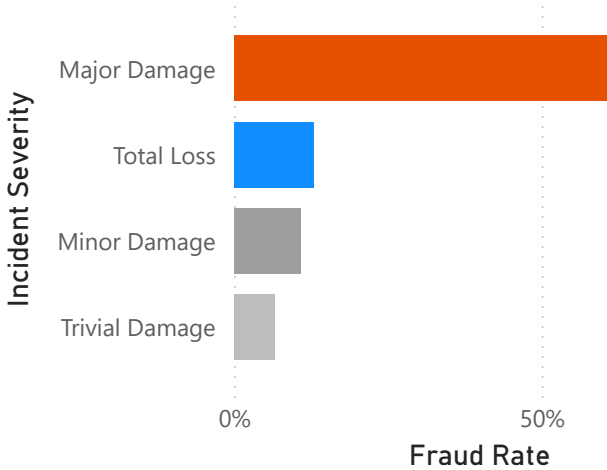
Claims by Risk Tier



Risk Tier



Fraud Rate by Incident Severity



Insurance Fraud Detection - Flagged Claims

claim_index	actual_fraud	fraud_probability	risk_tier	incident_severity_label	incident_type_label	total_claim_amount
35	Y	1.00	High	Major Damage	Single Vehicle Collision	53100
39	Y	1.00	High	Major Damage	Single Vehicle Collision	60170
66	Y	1.00	High	Minor Damage	Single Vehicle Collision	89700
89	Y	1.00	High	Major Damage	Multi-vehicle Collision	45120
150	N	1.00	High	Total Loss	Single Vehicle Collision	82720
155	Y	1.00	High	Major Damage	Single Vehicle Collision	37280
234	Y	1.00	High	Major Damage	Multi-vehicle Collision	53600
257	Y	1.00	High	Major Damage	Single Vehicle Collision	57690
259	Y	1.00	High	Major Damage	Multi-vehicle Collision	47790
306	Y	1.00	High	Major Damage	Single Vehicle Collision	52800
328	Y	1.00	High	Major Damage	Multi-vehicle Collision	51810
460	Y	1.00	High	Minor Damage	Single Vehicle Collision	76230
535	Y	1.00	High	Major Damage	Single Vehicle Collision	52400
538	Y	1.00	High	Minor Damage	Parked Car	4700
579	Y	1.00	High	Major Damage	Single Vehicle Collision	64200
643	Y	1.00	High	Major Damage	Single Vehicle Collision	51920
715	Y	1.00	High	Major Damage	Multi-vehicle Collision	55400
785	Y	1.00	High	Major Damage	Multi-vehicle Collision	50050
838	Y	1.00	High	Major Damage	Multi-vehicle Collision	51370
878	Y	1.00	High	Major Damage	Multi-vehicle Collision	54400
895	Y	1.00	High	Major Damage	Multi-vehicle Collision	55200
939	Y	1.00	High	Major Damage	Multi-vehicle Collision	61050
967	Y	1.00	High	Major Damage	Single Vehicle Collision	54200
972	Y	1.00	High	Major Damage	Single Vehicle Collision	55400
23	Y	0.94	High	Total Loss	Single Vehicle Collision	98160
27	Y	0.94	High	Minor Damage	Vehicle Theft	5720

Total

Risk Tier

High

Low

Medium

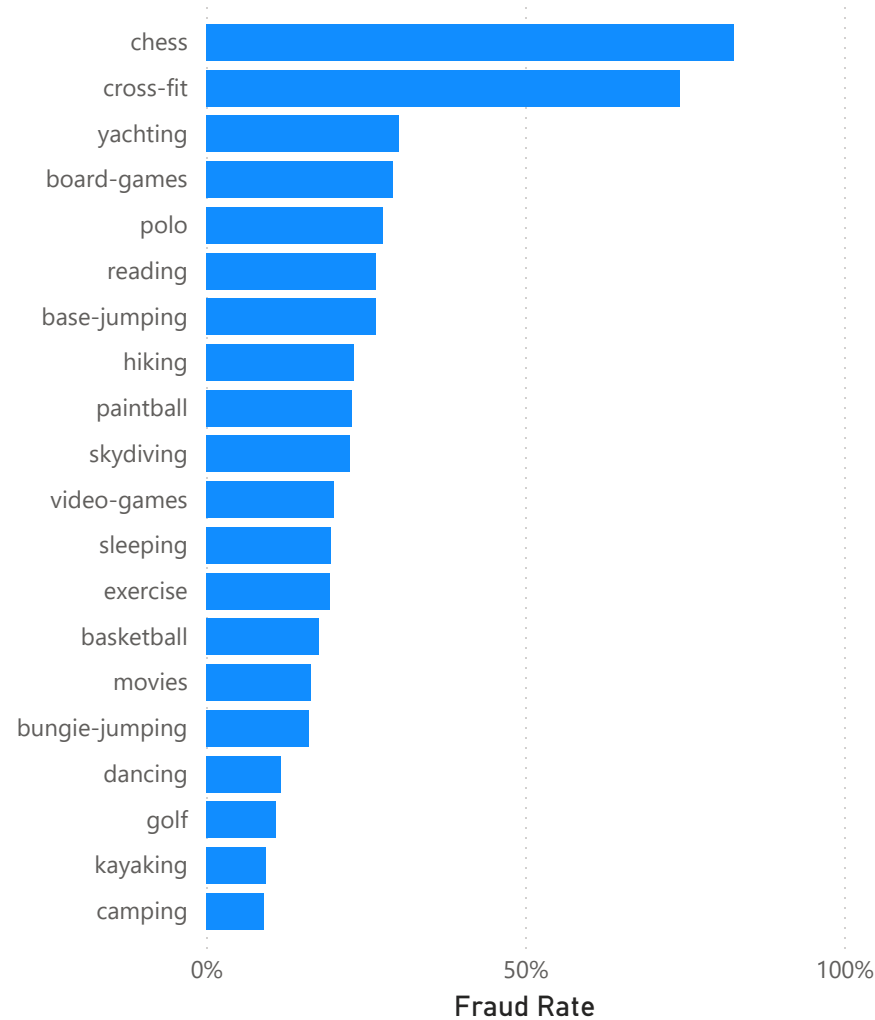
Fraud Confirmed

N

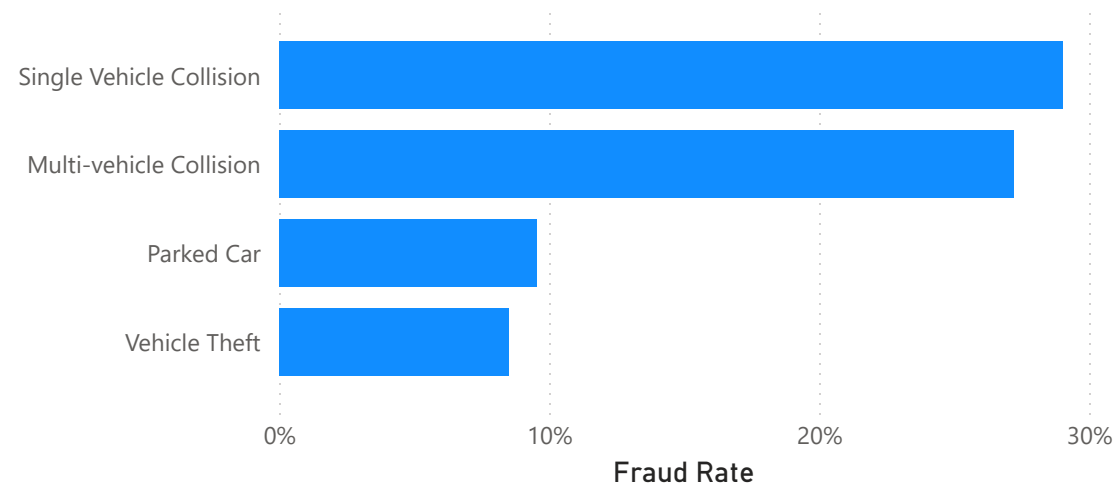
Y

Insurance Fraud Detection — Risk Factor Analysis

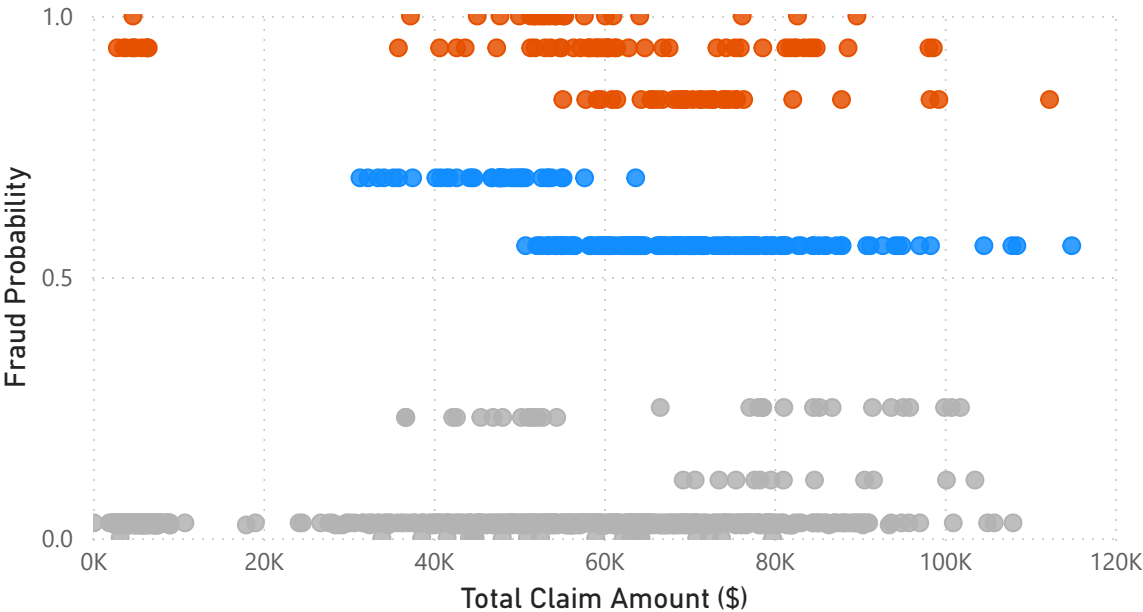
Fraud Rate by Insured Hobby



Fraud Rate by Incident Type



Claim Amount vs Fraud Probability by Risk Tier



Insurance Fraud Detection — Model Performance

71%

Model Recall

64%

F1 Score

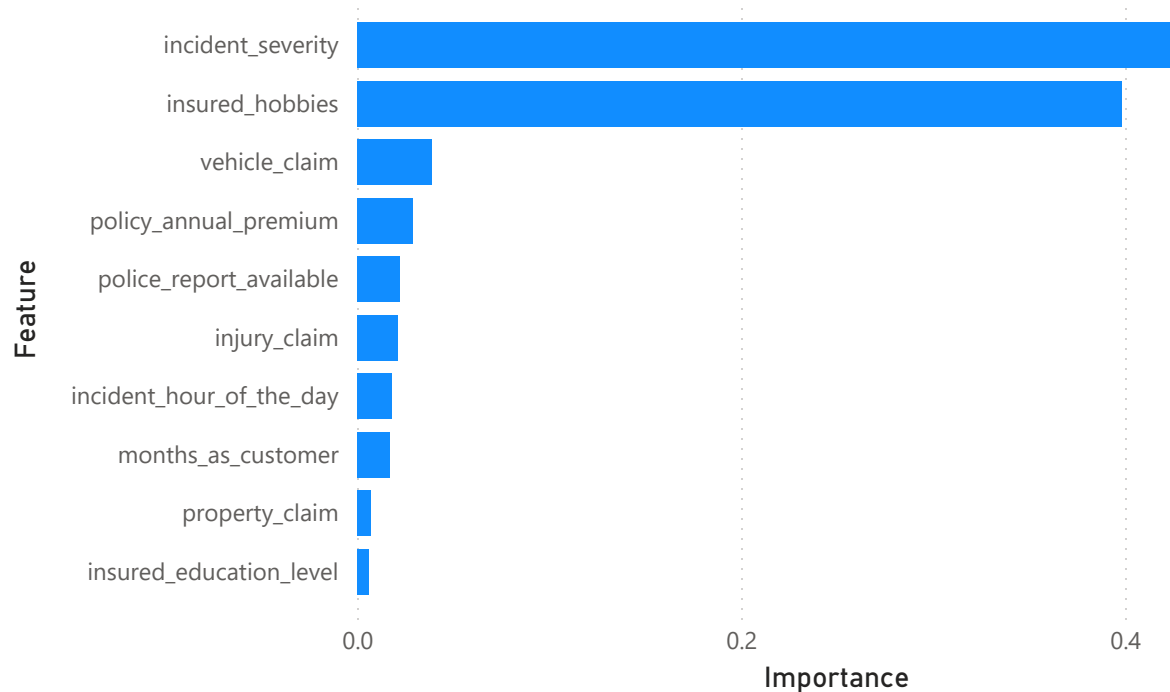
58%

Precision

0.763

ROC-AUC

Top 10 Feature Importance — Decision Tree



Actual \ Predicted	Not Fraud	Fraud
	▲	
Fraud	14	35
Not Fraud	126	25

Model Selection: Decision Tree (max_depth=5)

Two models were evaluated on 200 held-out test claims. The Decision Tree was selected over Logistic Regression based on superior fraud recall (71% vs 35%). In insurance fraud detection, missing a fraudulent claim (false negative) carries higher cost than investigating a legitimate one (false positive). The Decision Tree correctly identified 35 of 49 actual fraud cases in the test set. Incident severity and insured hobbies account for 83% of the model's predictive weight.